# EEVEE: Towards Test-time Prompt Learning in the Real World for Self-Improving Agents

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HarnessRSI 63 ★★

evidence: retained

Paper

Code

## 1 · What changes, what is kept

**Initialize complementary slots**
Mixed training Q&A → prompt search.
Choose 4 prompts by validation coverage.

**Evolve router · prompts fixed**
Mutation + misrouting analysis + reflection.
Signal: ground-truth correctness.
Keep only above phase-baseline validation.

**Evolve prompts · router fixed**
Route train / validation into slot groups.
Reflect on target + model answer + score.
Keep above empty prompt AND on Pareto front.

**Alternate, then converge**
Router and prompt phases inform each other.
Freeze router; larger per-slot prompt search.

**Final test: input → learned router → slot → answer**
Weights fixed. Test examples do not update prompts.
Prepared labeled adaptation, not fully online learning.

**Budget:** 10 bootstrap steps; 150 co-evolution mini-steps; final 60 steps per label, mini-batch 5. Validation gates do not guarantee no regressions.

## 2 · Main macro scores, not best seeds

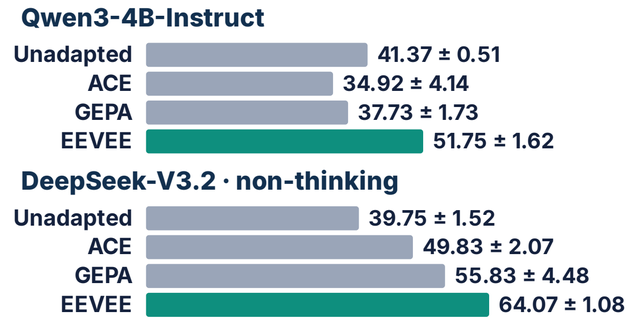


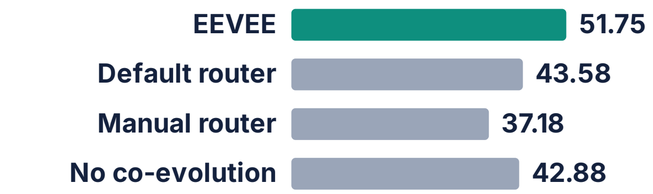

Table 7: mean ± sample SD, three independent runs. Four-benchmark macro average (%). Per-task SD and CIs are not reported.

## 4 · Where the gain is absent or thin

- **HumanEval leads:** Qwen3-4B-Instruct 49.46 → 72.63; DeepSeek-V3.2 42.82 → 92.82.
- **GPQA Diamond declines:** Qwen 56.00 → 54.55; DeepSeek 64.98 → 63.08 (Table 1).
- **DeepSeek Formula:** EEVEE 60.55 vs GEPA 60.56; near-tie without per-task variance.
- **Unseen tasks (Table 3):** MBPP 69.29 → 70.42; MMLU-Pro 70.74 → 68.92. Combined 70.01 → 69.67; means without SD/CI.
- GPQA retest (Table 4): −3.7 / −7.7 points; 1/6 positive runs. Separate from Table 1.

## 3 · Controls, protocol & cost

**Ablation · Qwen3-4B-Instruct (Table 2)**




Manual: written once by GPT-5.4. No row SD or explicit search-run count; no equal-compute assurance. Default-router TheoremQA 26.10 vs full 25.27.

**Held-out data, not a transductive final test**

Cap 500 per benchmark; half test, half adaptation; half of adaptation reserved for validation. Train/validation excluded from test (Appendix B).

Validation selects candidates; the split for choosing hyperparameter defaults is unspecified. GEPA / ACE access parity and tuning/search budgets are not itemized; neither gets an EEVEE-router control.

**Transfer and inference cost**

Qwen-learned prompts on DeepSeek-V3.2: 39.75 → 54.10; router-transfer details unspecified (Table 3).

Qwen final tokens / test example
**EEVEE 4.32k · GEPA 3.47k · ACE 21.30k**

Adds routing before answering. These exclude evolution cost; total adaptation tokens / dollars and matched search budgets are not reported. No separate contamination audit.

## Takeaway

Co-evolving a router and specialized prompts helps mixed, labeled adaptation—especially executable and formula-based procedures. Knowledge QA can regress. Evidence supports retained adaptation, not fully online learning or a compute-matched win.

## Our read · HarnessRSI

**Score 63 · ★★ · retained**

**Reports:** held-out splits, transfer, ablations, all-method macro SD (Table 7).
**Missing:** search cost, matched budgets, tuning split. Stored variance judgment incomplete.